

When Does Risk-Managed Momentum Add Value?

An Out-of-Sample, Regime-Conditional, and Statistically Tested Study
of a Volatility-Targeted Dual-Momentum Strategy

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Code, cached data, and reproducibility instructions are available at github.com/aarivgoyal/quant-momentum-risk-targeting.
Educational research only, not investment advice. All results are reproducible from the cached dataset and a fixed seed.

Abstract

This project asks a sharper question than whether a systematic strategy can “beat the market.” A defensive, risk-managed strategy is expected to trail a passive equity benchmark across a sustained bull market, so demonstrating that is not a finding. The meaningful question is conditional: under what market conditions does risk-managed allocation actually add value, and are any differences statistically distinguishable from noise?

I build a rules-based backtesting framework that combines dual-momentum asset selection with volatility targeting across four asset classes (U.S. equities, international equities, aggregate bonds, and short-term Treasuries), and subject it to a battery of tests: a corrected backtest methodology that eliminates lookahead bias, charges transaction costs on portfolio turnover, and evaluates risk-adjusted return against a real time-varying risk-free rate; an in-sample/out-of-sample split and a walk-forward parameter-selection procedure that test for overfitting; a regime-conditional analysis isolating crisis windows and up- versus down-market months; a sensitivity grid across the full parameter space; and a block-bootstrap test of statistical significance.

The results are instructive precisely because they are mixed. Over the full sample (Feb 2013 - Jun 2026), the strategy reduces volatility (12.4% vs. 17.0%) and equity drawdown (−28.8% vs. −34.0%) relative to buy-and-hold, but it does not outperform a simple 60/40 portfolio on any major metric, and block-bootstrap testing shows that none of its risk-reduction benefits reach statistical significance: its only statistically reliable effect is sacrificing return relative to pure equity (CAGR −6.5 points, $p = 0.011$). The value of the strategy is regime-dependent: in the slow 2022 bear market it outperformed both benchmarks (−14.2% vs. −24.1% and −20.2%), yet in the fast COVID crash it suffered its worst drawdown because a monthly momentum filter reacts too slowly. Walk-forward re-optimization produced worse out-of-sample results than fixed parameters, an empirical demonstration of the backtest-overfitting trap. Finally, motivated directly by the diagnosed fast-crash weakness, an engineered fix (a daily crash-guard overlay) sharply reduces fast-crash losses, improving the COVID crisis return from -28.5% to -1.9% (and its maximum drawdown from -28.8% to -4.0%), with the 2018 selloff improving comparably. It delivers statistically significant downside protection against both benchmarks in down markets; yet it cannot improve full-sample risk-adjusted return, because the same speed that protects against crashes causes whipsaw losses in calm markets, exposing a fundamental crash-protection-versus-whipsaw tradeoff. The contribution is methodological as much as empirical: it shows how a financial idea can be specified mathematically, implemented without common backtest errors, iterated through a diagnosed weakness, and judged honestly enough to reveal exactly where it works and where it does not.

1. Introduction

Financial markets are dynamic systems shaped by economic forces, shifting expectations, and uncertainty. Because prices respond both to information and to human behavior, outcomes are often influenced by emotion and short-term reaction. That makes markets a natural setting for quantitative modeling: rather than relying on judgment, a quantitative framework transforms noisy price data into explicit, testable decision rules.

Two ideas anchor this project. Momentum is the empirical tendency of assets with strong recent performance to continue outperforming over intermediate horizons. Risk management, here through volatility-based position sizing, reduces exposure when market instability rises. Combined, they recast portfolio allocation as a mathematical decision problem: generate a signal, classify the regime, size the position, and evaluate the result through simulation.

A naive version of this project would build such a strategy, compare its terminal wealth to a benchmark, and declare success or failure. That framing is weak, because the answer is largely predetermined by the sample: over 2013-2026, U.S. equities delivered one of the strongest bull runs in history, so any strategy that holds cash or bonds part of the time will trail buy-and-hold over that window almost by construction. The scientific question is therefore not “does it win?” but “when does it add value, and can we tell that value from luck?” This paper is organized around that question, and the methodology is built to answer it credibly rather than flatteringly.

2. Literature Review

Cross-sectional momentum. Jegadeesh and Titman (1993) documented that assets ranked on trailing returns continue to outperform or underperform over 3-12 month horizons, establishing momentum as a robust, systematic signal. A common refinement skips the most recent month of the formation window to avoid short-term reversal (the “12-1” convention), which this project tests in its extension model.

Time-series (absolute) momentum. Moskowitz, Ooi, and Pedersen (2012) showed that an asset’s own past return predicts its future return, motivating an absolute-momentum regime filter that decides whether to take risk at all, independent of cross-sectional ranking.

Dual momentum. Antonacci (2014) combined absolute and relative momentum into one framework: relative momentum selects the strongest asset within a risk-on universe, while absolute momentum decides whether to hold risk assets or retreat to defensives. This project’s decision rule is a direct implementation of that two-layer logic.

Volatility management. Moreira and Muir (2017) found that scaling exposure inversely to recent volatility can improve risk-adjusted returns, since volatility is persistent and weakly compensated in the short run. This is the theoretical basis for the volatility-targeting overlay.

Backtest overfitting. A model that looks excellent under one parameter set may simply be fit to historical noise. Bailey, Borwein, López de Prado, and Zhu (2014) and Harvey and Liu (2015) formalize how easily in-sample performance overstates out-of-sample reality and argue for out-of-sample validation and significance testing rather than single-path comparisons. These concerns directly motivate Sections 6-9, and, as Section 6.2 shows, they are vindicated empirically here.

3. Research Question and Hypotheses

Research question. Under what conditions does a dual-momentum strategy with volatility targeting improve risk-adjusted performance relative to passive benchmarks, and are the differences statistically meaningful?

This decomposes into four falsifiable hypotheses:

- *H1 - Downside protection.* Across the full sample, the strategy exhibits lower volatility and a shallower maximum drawdown than buy-and-hold equity.
- *H2 - Conditional value.* The strategy’s relative performance is strongest in equity-stress regimes and weakest in calm bull markets: it earns its keep when downside protection matters.
- *H3 - Robustness.* Performance is stable across plausible parameter choices and persists out-of-sample, rather than being an artifact of one lucky configuration.
- *H4 - Significance.* Observed differences in risk-adjusted performance versus benchmarks are statistically distinguishable from zero under a block-bootstrap test.

Stating the question as falsifiable hypotheses, rather than “can I beat the market?”, is what allows an honest, mixed result to remain scientifically informative.

4. Methodology

4.1 Data and Asset Universe

Daily, dividend-adjusted price data is obtained through `yfinance` for four ETFs: VOO (U.S. equities), VXUS (international equities), BND (U.S. aggregate bonds), and SHY (short-term Treasuries, the cash-like allocation). The data spans 2012-01-03 to 2026-06-15. To ensure reproducibility, prices are downloaded once and cached; all runs read the pinned file, so reported figures do not drift as the vendor revises history. The daily return of asset i on day t is

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1.$$

4.2 Momentum Signal

For a lookback of L trading days and an optional skip of s days,

$$M_{i,t}(L, s) = \frac{P_{i,t-s}}{P_{i,t-s-L}} - 1.$$

The baseline uses $L = 252$ ($\approx$ one year), $s = 0$; the extension uses $s = 21$ (“12-1”). Momentum is applied in two layers:

1. *Absolute-momentum regime filter.* If $M_{\text{VOO},t} > 0$, the market is risk-on; otherwise risk-off.
2. *Relative-momentum selection (risk-on).* Among $\mathcal{U} = \{\text{VOO}, \text{VXUS}\}$, choose $j_t = \arg \max_{i \in \mathcal{U}} M_{i,t}$.

In a risk-off regime, the strategy allocates to the defensive asset (BND).

4.3 Volatility Estimation and Targeting

Realized annualized volatility over a window K (baseline $K = 20$) is

$$\sigma_{i,t}(K) = \sqrt{252} \text{Std}(r_{i,t-K+1}, \dots, r_{i,t}).$$

The selected asset is sized so its ex-ante contribution targets an annual volatility $\sigma^* = 10\%$:

$$w_t = \text{clip}\left(\frac{\sigma^*}{\sigma_{j,t}(K)}, 0, 1\right), \quad w_{\text{cash},t} = 1 - w_t.$$

The cap $w_{\max} = 1$ forbids leverage and shorting, making the rule deliberately asymmetrically conservative: it can step down risk when volatility is high but cannot lever up when volatility is low. As Section 5 shows, this asymmetry is a meaningful drag in a bull market.

4.4 Portfolio Dynamics, Timing, and Costs

A decision computed at month-end t is applied starting the next trading day, so no decision uses information from the bar whose return it earns (eliminating the lookahead present in the initial implementation). Transaction costs are charged on turnover (any change in target weights, including the monthly volatility re-sizing), not only when the held asset changes:

$$c_t = \kappa \sum_i |w_{i,t} - w_{i,t-1}|, \quad \kappa = 0.001, \quad r_{p,t} = \sum_i w_{i,t} r_{i,t} - c_t,$$

$$V_t = V_{t-1}(1 + r_{p,t}), \quad V_0 = \$10,000.$$

This stricter cost model is one reason the corrected results are more conservative than the initial implementation.

4.5 Benchmarks

Buy-and-hold VOO (passive equity) and a 60/40 VOO/BND portfolio rebalanced monthly, rather than continuously and costlessly, so the comparison matches the strategy's own cadence.

4.6 Performance Metrics

With n trading days and daily risk-free rate r_t^f :

$$\text{CAGR} = \left(\frac{V_T}{V_0}\right)^{252/n} - 1, \quad \sigma_p = \sqrt{252} \text{Std}(r_{p,t}),$$

$$\text{Sharpe} = \sqrt{252} \frac{\text{mean}(r_{p,t} - r_t^f)}{\text{Std}(r_{p,t} - r_t^f)}, \quad \text{Sortino} = \sqrt{252} \frac{\text{mean}(r_{p,t} - r_t^f)}{\text{Std}_{\text{downside}}},$$

$$\text{MaxDD} = \min_t \left(\frac{V_t}{\max_{s \leq t} V_s} - 1 \right), \quad \text{Calmar} = \frac{\text{CAGR}}{|\text{MaxDD}|}.$$

The risk-free rate is a real, time-varying daily series (from the BIL T-bill ETF), not $r^f = 0$. This matters: the sample spans the near-zero-rate 2010s and the 4-5% environment of 2022-2024, and setting $r^f = 0$ mechanically inflates every Sharpe ratio while unfairly crediting a strategy that parks capital in cash during defensive periods. Correcting this is a primary reason the strategy's Sharpe falls from 0.75 (initial implementation) to 0.58 here.

5. Baseline Results (Full Sample, Feb 2013 - Jun 2026)

Table 1. Full-sample performance of the risk-targeted strategy against both benchmarks.

Portfolio	CAGR	Ann. Vol	Sharpe	Sortino	MaxDD	Calmar	Final (\$)
Risk-Targeted Dual Momentum	8.30%	12.37%	0.58	0.69	−28.79%	0.29	\$29,299
Buy & Hold VOO	14.75%	16.95%	0.81	0.98	−33.99%	0.43	\$63,539
60/40 (monthly)	9.84%	10.57%	0.79	0.96	−21.95%	0.45	\$35,231

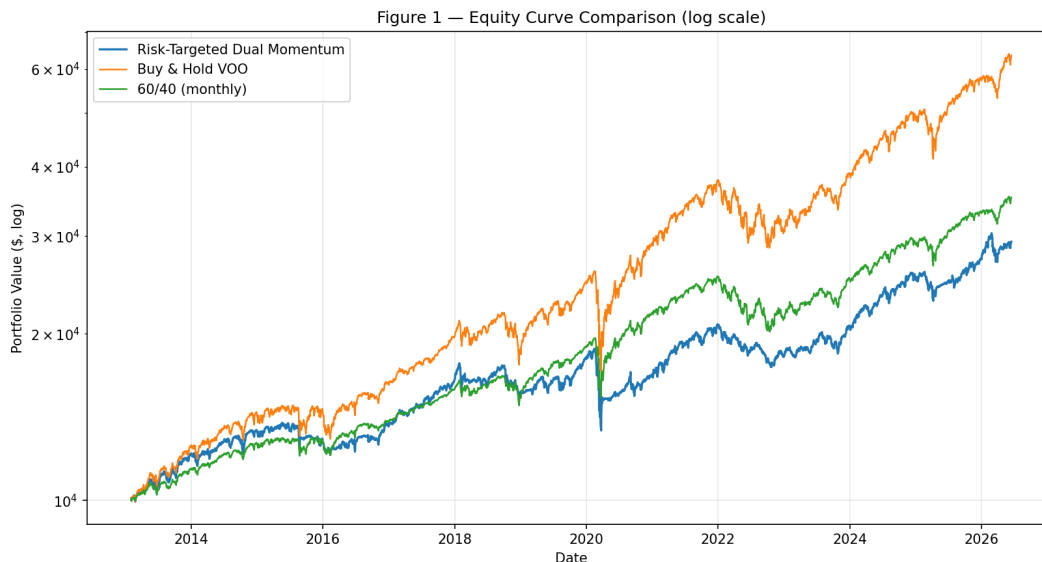


Figure 1. Equity-curve comparison, log scale.

Interpretation. H1 is supported: the strategy’s volatility (12.4%) and maximum drawdown (−28.8%) are both lower than buy-and-hold VOO’s (17.0% and −34.0%), exactly as volatility targeting and defensive rotation intend. However, the strategy is dominated by the 60/40 benchmark on *every* metric: lower CAGR (8.30% vs. 9.84%), lower Sharpe (0.58 vs. 0.79), and a deeper drawdown (−28.8% vs. −22.0%). And versus VOO it converts risk into return less efficiently, not more: a lower Sharpe (0.58 vs. 0.81) means the volatility it removed cost more return than it was worth over this sample. Read in isolation, the full-sample average says the strategy is an inefficient risk-reducer. The remainder of the paper asks whether that average conceals a more useful conditional truth.

6. Out-of-Sample and Walk-Forward Validation

6.1 In-Sample / Out-of-Sample Split (at 2020-01-01)

Because the baseline rule has no fitted parameters, this tests whether its behavior is stable across two very different macro regimes: the 2013-2019 low-volatility bull versus the 2020-2026 period of a pandemic crash, an inflation shock, and a rate-hiking cycle.

The strategy’s relative ranking is consistent across both periods (it protects against equity drawdown but trails on risk-adjusted return in both), so its character is not a single-regime artifact. Notably, every portfolio’s

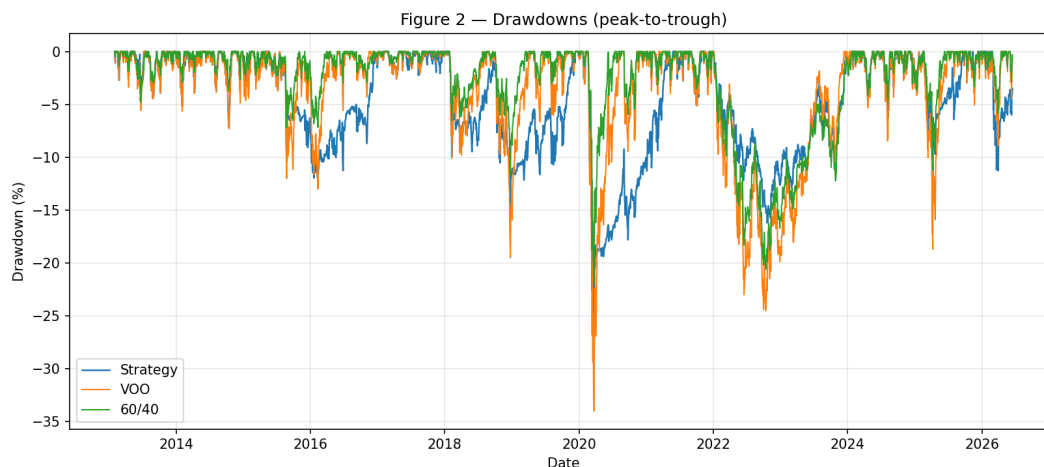


Figure 2. Drawdown comparison, peak-to-trough.

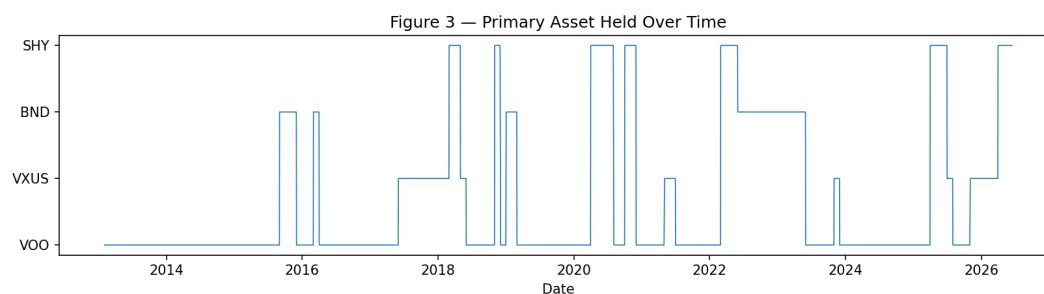


Figure 3. Primary asset held over time.

Sharpe falls out-of-sample as volatility rises, but the strategy's deepest drawdown (-28.8%) occurs entirely in the out-of-sample period, foreshadowing the COVID result in Section 7.

6.2 Walk-Forward Parameter Selection: the Overfitting Test

The procedure rolls through time: on each trailing 4-year window it selects the parameter set with the best in-sample Sharpe, applies only that choice to the next year, and stitches the out-of-sample pieces into one track using no future information.

This is the most important methodological result in the paper. Re-optimizing parameters each year (the intuitive thing to do) produced a track that was worse on every metric than simply fixing sensible parameters (Sharpe 0.33 vs. 0.58), and the “optimal” parameters were wildly unstable from year to year. This is a clean, empirical demonstration of the backtest-overfitting trap warned about by Bailey et al. (2014) and Harvey & Liu (2015): chasing the best in-sample fit selects noise, not signal, and degrades out-of-sample performance. The practical lesson (favor robust fixed parameters over optimization) is one of the paper's genuine contributions.

7. Regime-Conditional Analysis

This section tests H2: a defensive strategy should justify its drag by protecting capital when equities fall.

Table 2. In-sample and out-of-sample performance, split at 2020-01-01.

Period	Portfolio	CAGR	Ann. Vol	Sharpe	MaxDD
In-Sample (2013-2019)	Strategy	8.75%	10.28%	0.82	−14.56%
	VOO	13.81%	12.86%	1.03	−19.48%
	60/40	9.59%	7.55%	1.18	−11.59%
Out-of-Sample (2020-2026)	Strategy	7.68%	14.28%	0.41	−28.79%
	VOO	15.63%	20.45%	0.69	−33.99%
	60/40	10.03%	13.06%	0.60	−21.95%

Table 3. Walk-forward re-optimization versus fixed baseline parameters.

Walk-forward result	Value
Walk-forward CAGR / Sharpe / MaxDD	5.51% / 0.33 / −28.79%
Fixed-baseline CAGR / Sharpe / MaxDD (for comparison)	8.30% / 0.58 / −28.79%
Selected configurations over time	Highly unstable, e.g., lookback jumped 252 → 504 → 63 → 504; target vol cycled 5% → 10% → 15% across consecutive years

7.1 Crisis Windows

Table 4. Performance within three equity-stress windows.

Window	Strategy		VOO		60/40	
	Ret	MaxDD	Ret	MaxDD	Ret	MaxDD
2018 Q4 Selloff	−13.29%	−13.67%	−19.03%	−19.39%	−11.36%	−11.55%
COVID Crash (Feb-Mar 2020)	−28.50%	−28.79%	−33.68%	−33.99%	−21.72%	−21.95%
2022 Bear Market	−14.23%	−14.60%	−24.06%	−24.52%	−20.18%	−20.26%

The crisis windows reveal a sharp, non-obvious pattern that the full-sample average completely hides:

- In the 2022 bear market, the strategy outperformed both benchmarks decisively, losing 14.2% against VOO’s 24.1% and the 60/40’s 20.2%. 2022 was one of the worst years on record for balanced portfolios because stocks and bonds fell together; the strategy’s absolute-momentum filter had time to rotate it defensive, and it protected roughly 10 points of capital versus equities and 6 points versus the 60/40. This is the strongest evidence for H2.
- In the COVID crash, the strategy failed at its core purpose: its worst drawdown of the entire sample (−28.8%) occurred here, because the crash unfolded in roughly one month, faster than a monthly-rebalanced momentum filter can react. It still beat VOO by ~5 points but lost badly to the 60/40.

The reconciliation is the paper’s central insight: risk-managed momentum protects against slow bear markets but not fast crashes. The mechanism that helps in 2022 is the same mechanism that fails in 2020 (a monthly signal), and recognizing that the strategy’s value is conditional on the speed of a drawdown, not merely its presence, is what separates a real finding from a marketing claim.

7.2 Up- versus Down-Market Conditioning

Monthly returns split by whether VOO rose or fell that month:

Table 5. Average monthly returns conditioned on the direction of VOO.

	VOO Up Months ($n = 112$)	VOO Down Months ($n = 49$)
Strategy avg monthly return	+2.10%	−2.46%
VOO avg monthly return	+3.34%	−3.57%

The strategy captures about 63% of VOO’s up-month gains but participates in roughly 69% of its down-month losses, an unfavorable average capture ratio, since it gives up more relative upside than it spares on the downside. Crucially, this average is dragged down by the COVID months: the conditional protection that appears cleanly in the 2022 window does not show up in the all-months average, precisely because the fast-crash failures offset the slow-bear successes. H2 is therefore partially supported: conditional value exists, but only in a specific regime, not as a blanket down-market benefit.

8. Parameter Sensitivity

Robustness (H3) requires that conclusions not hinge on one arbitrary calibration. The strategy is re-run across 36 configurations: lookbacks $\{63, 126, 252, 504\}$, volatility windows $\{20, 60, 90\}$, target volatilities $\{5\%, 10\%, 15\%\}$.

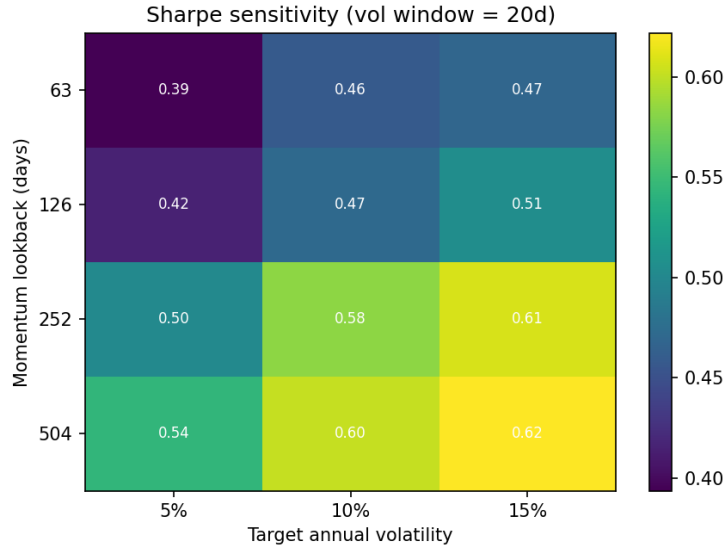


Figure 4. Sharpe heatmap across lookback and target volatility.

Two honest conclusions follow. First, the baseline was *not* cherry-picked: it sits in the upper third but is not the maximum, which strengthens the claim that the reported result is representative rather than curve-fit. Second, and more decisively, no parameter setting anywhere in the grid beats the simple 60/40 on a risk-adjusted basis. The strategy’s underperformance versus a balanced benchmark is therefore *structural*, not a tuning failure, exactly the kind of robust statement that significance testing in Section 9 then formalizes. The

Table 6. Summary of the 36-configuration sensitivity sweep.

Sensitivity finding	Value
Range of Sharpe across all 36 configurations	0.32 - 0.62
Best configuration	$L = 504$, $K = 20$, $\sigma^* = 15\%$ (Sharpe 0.62)
Baseline (252/20/10%) rank	8th of 36 (Sharpe 0.58)
Configurations beating the 60/40 benchmark (Sharpe 0.79)	0 of 36

response surface is also reasonably smooth (higher target volatility and longer lookbacks help monotonically), indicating the behavior is structural rather than a fragile spike.

9. Statistical Significance

Point-estimate differences may be noise. Using a stationary block bootstrap (block length 21 days, to preserve short-horizon autocorrelation), the strategy and a benchmark series are jointly resampled 2,000 times; the metric difference (strategy – benchmark) is recomputed each time to form a sampling distribution, yielding a 95% CI and a two-sided bootstrap p -value for H_0 : difference = 0.

Table 7. Block-bootstrap significance tests over the full sample.

Comparison	Metric	Point diff	95% CI	p -value
Strategy – 60/40	Sharpe	–0.21	[–0.48, +0.06]	0.117
	CAGR	–1.5%	[–4.9%, +1.7%]	0.372
	MaxDD	–6.8%	[–13.6%, +2.7%]	0.185
Strategy – VOO	Sharpe	–0.22	[–0.47, +0.02]	0.083
	CAGR	–6.5%	[–11.9%, –1.7%]	0.011
	MaxDD	+5.2%	[–2.3%, +14.0%]	0.126

Interpretation (H4). This is where intellectual honesty becomes the result. Versus the 60/40, none of the differences are statistically significant (every confidence interval includes zero), meaning the strategy is statistically indistinguishable from a balanced benchmark over this sample, despite point estimates that favor the 60/40. Versus VOO, the only statistically significant difference is that the strategy gives up return (CAGR –6.5 points, $p = 0.011$), the expected cost of being defensive in a bull market. Critically, the strategy’s apparent benefits (lower drawdown and lower volatility) do not reach significance (the MaxDD-reduction CI spans zero, $p = 0.126$). So the rigorous verdict is that the strategy’s downside protection, while visible in point estimates and decisive within the 2022 window, is not statistically demonstrable at the full-sample level. Most retail backtests never run this test, and many would quietly fail it.

9.1 Conditional (Down-Market) Significance

A full-sample bootstrap is a blunt instrument for a strategy whose benefits are concentrated in stress periods: calm-market noise drowns out the crisis signal. A sharper test restricts to the 49 months in which VOO fell and bootstraps the mean monthly difference (strategy – benchmark):

Conditioning on down markets sharpens the picture considerably: the strategy’s downside protection versus

Table 8. Down-market-only significance, baseline strategy.

Down-market comparison	Mean monthly diff	95% CI	p-value
Strategy – VOO	+1.11%	[+0.61%, +1.65%]	< 0.001
Strategy – 60/40	–0.17%	[–0.64%, +0.31%]	0.47

pure equity is statistically real (it loses about 1.1% less per down-month than VOO, with the interval comfortably above zero), even though the full-sample test could not establish it. Against the 60/40, however, the strategy remains statistically indistinguishable in stress months. That residual gap (significant protection versus equity, none versus a balanced benchmark) is exactly what the engineered fix in Section 11 sets out to close.

10. Extension: Diversified Basket and Skip-Month Momentum

The baseline’s two most-criticized design choices (holding only a single risk-on asset, and using plain 12-month momentum without the academic skip) are addressed in an extension that (i) holds a volatility-scaled equal-weight basket of all positive-momentum risk-on assets and (ii) uses 12-1 skip-month momentum for selection.

Table 9. Baseline versus the diversified, skip-month extension.

Portfolio	CAGR	Ann. Vol	Sharpe	MaxDD
Baseline (single, no skip)	8.30%	12.37%	0.58	–28.79%
Extension (basket, 12-1)	5.61%	11.83%	0.38	–29.95%

Interpretation. Contrary to the hypothesis, the principled refinements made performance worse, not better: the extension’s Sharpe fell to 0.38 with a slightly deeper drawdown. Two plausible explanations: over this sample, concentrating in the single strongest-momentum asset happened to beat diversifying across a weaker one (VXUS lagged VOO for most of the period, so equal-weighting dragged returns down), and the skip-month convention reduced the signal’s responsiveness without a compensating reduction in whipsaw. This is a genuine negative result for the extension, but it is still a finding: it shows that not every theoretically-motivated improvement survives empirical testing, and it constrains the space of refinements worth pursuing. Demonstrating that iteration honestly (including when the iteration fails) is the defining habit of real research.

11. Engineering a Fix: A Fast Crash-Guard Overlay (v3)

The regime analysis (Section 7) localized the strategy’s single worst failure: a once-a-month signal cannot react to a crash that unfolds in days, which is why the COVID drawdown (–28.8%) was the deepest in the sample. A naive response would be to tune parameters until the backtest improved, but Section 6.2 showed that approach backfires. Instead, this section tests a fix motivated directly by the diagnosed mechanism: a daily crash-guard overlay layered on top of the monthly strategy.

The fix. Each trading day, a fast 20-day trend gauge is computed on U.S. equities. When that fast trend is negative, the guard moves the equity sleeve (VOO/VXUS) entirely to cash until the trend recovers; bonds and cash are never gated. The guard is applied with a one-day delay, so it never uses information from the day it

trades on. This lets the strategy step out of equities within days of a downturn beginning, rather than waiting up to a month for the next rebalance. The result is genuinely uncertain in advance: faster reaction should help in crashes but risks “whipsaw”: repeatedly cutting and re-entering in choppy markets.

Table 10. Full-sample performance: unguarded v2 versus the crash-guarded v3.

Portfolio	CAGR	Ann. Vol	Sharpe	MaxDD	Final (\$)
v2 Baseline (no guard)	8.30%	12.37%	0.58	−28.79%	\$29,299
v3 (+ crash guard, 20-day)	2.60%	7.61%	0.17	−22.24%	\$14,230

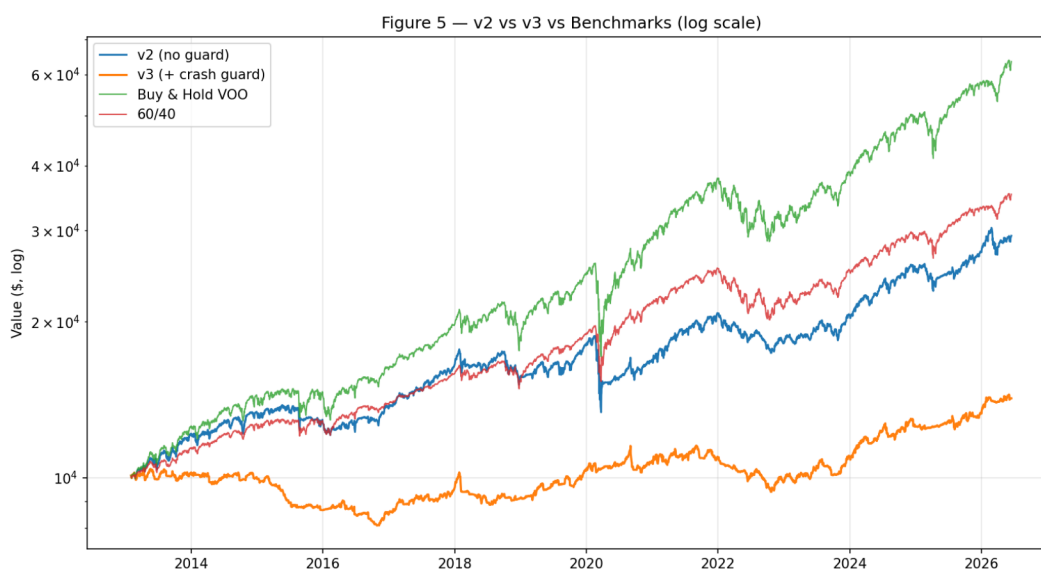


Figure 5. Full-sample equity curves, v2 vs. v3 vs. benchmarks, log scale.

The full-sample numbers look like a step backward, and on risk-adjusted return, they are: the guard’s constant cutting and re-entering bled so much return that the Sharpe collapsed from 0.58 to 0.17. But the full-sample average hides what the guard was *built* to do. The crisis windows tell the real story:

Table 11. Crisis-window returns and drawdowns, v2 versus v3.

Window	v2		v3	
	Return	MaxDD	Return	MaxDD
2018 Q4 Selloff	−13.29%	−13.67%	−1.57%	−2.96%
COVID Crash (Feb-Mar 2020)	−28.50%	−28.79%	−1.94%	−3.97%
2022 Bear	−14.23%	−14.60%	−12.35%	−12.73%

The guard does precisely what it was designed to do. In the two *fast* selloffs (2018 Q4 and COVID), it sharply reduced the loss, cutting the COVID return from −28.5% to −1.9%. In the slow 2022 bear it helped only marginally, because the monthly signal already handled a gradual decline. And the conditional significance test now strengthens markedly:

The guard closes the gap that v2 could not (Section 9.1): v3 now loses statistically less than both benchmarks

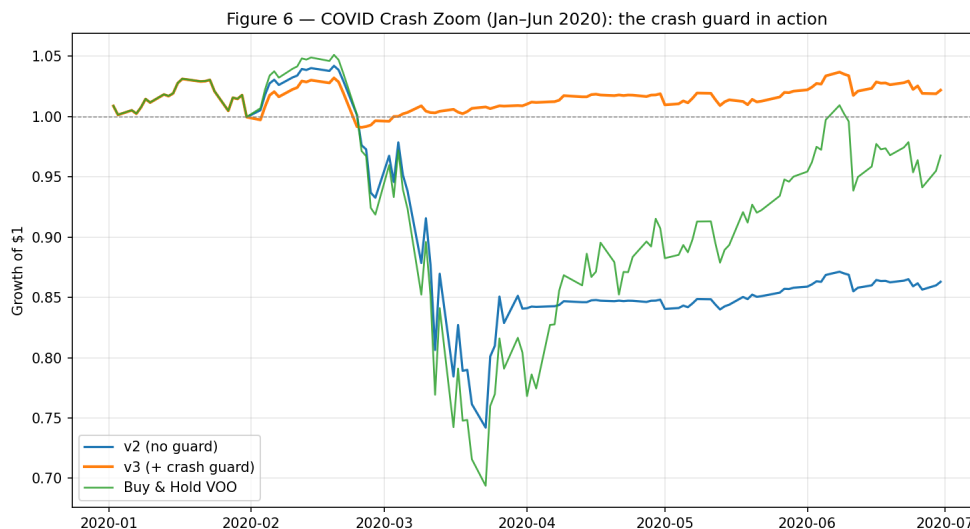


Figure 6. COVID-crash zoom (Jan–Jun 2020): the guard steps out of equities as the fast trend breaks, sidestepping almost the entire drawdown.

Table 12. Down-market significance for the crash-guarded v3 strategy.

Down-market comparison (v3)	Mean monthly diff	95% CI	<i>p</i> -value
v3 – VOO	+1.99%	[+1.10%, +2.94%]	< 0.001
v3 – 60/40	+0.70%	[+0.02%, +1.39%]	0.045

in down markets, including the 60/40, against which the unguarded strategy was indistinguishable.

The fundamental tradeoff. Why, then, does the guard hurt overall? Because crash protection and whipsaw are two ends of the same speed dial. Re-running the guard across reaction speeds makes the tension explicit:

Table 13. Guard reaction speed versus the crash-protection/whipsaw tradeoff.

Guard speed (fast-days)	Sharpe	CAGR	MaxDD
10	−0.24	−0.4%	−26.97%
20 (reported)	0.17	2.6%	−22.24%
40	0.34	4.2%	−21.93%
60	0.58	6.7%	−16.05%

Faster guards protect best in sudden crashes but whipsaw most in calm markets, destroying return; slower guards whipsaw less but react less to the fastest crashes. No setting beats v2’s full-sample Sharpe, but note the 60-day guard matches v2’s Sharpe (0.58) while roughly halving the worst drawdown (−16.0% vs. −28.8%), a genuine if modest improvement in tail protection at equal risk-adjusted return.

Interpretation. The fix succeeded at its stated objective and produced a statistically significant improvement in conditional downside protection, yet it could not raise full-sample risk-adjusted return, because reacting faster is inseparable from whipsawing more. This is not a failure of the fix; it is a discovery about the structure of the problem. Crash insurance has a premium, and a daily trend filter pays that premium continuously in

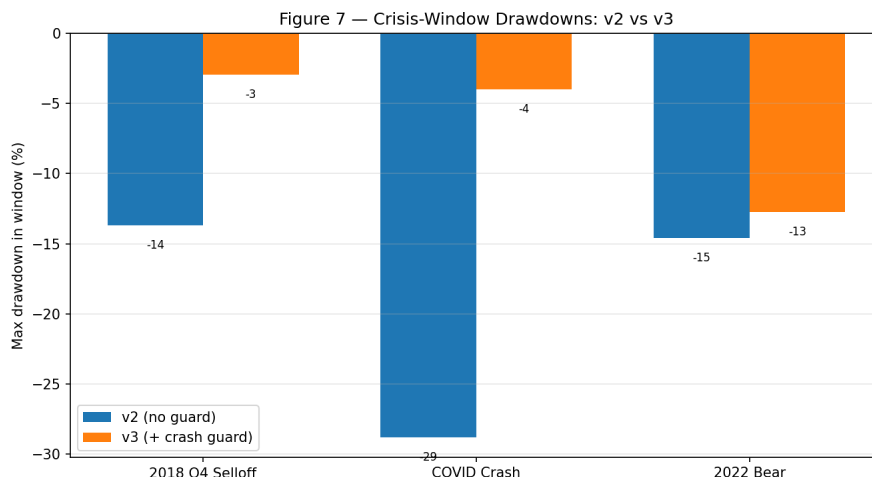


Figure 7. Maximum drawdown within each crisis window, v2 vs. v3.

exchange for protection that only matters occasionally. That tradeoff (the same one that defines real-world tail-hedging and trend-following strategies) is the deepest result of the entire project, and it only became visible because the strategy was diagnosed, fixed, and re-tested rather than declared finished.

12. Discussion

The full-sample comparison and the conditional analysis tell two different stories, and the gap between them is the point of the paper. On a full-sample average, the strategy is an inefficient risk-reducer: it trails a 60/40 portfolio on every metric, and significance testing cannot distinguish its defensive benefits from noise. Reading that average in isolation would label a partly-working risk-management system a failure. The regime analysis decomposes the average and shows that the strategy's protective value is real but conditional on the speed of the drawdown: it outperformed both benchmarks in the slow 2022 bear yet suffered its worst loss in the fast COVID crash, because the same monthly signal that protects gradually cannot react abruptly.

Several structural features explain the behavior. The strategy is concentrated (one risk-on asset at a time in the baseline); its volatility cap is asymmetrically conservative, surrendering upside in calm bull markets while only capping, never amplifying, exposure; and its single-asset defensive sleeve (BND) carries interest-rate risk that became a liability in 2022 even as the equity rotation helped. The walk-forward result adds a methodological lesson independent of the strategy itself: parameter optimization actively harmed out-of-sample performance, vindicating the literature's warnings about overfitting and arguing for robust, fixed rules. Each of these is a concrete, testable avenue for future work: a faster (e.g., dual-timescale) signal to address the crash-speed problem, a multi-asset defensive sleeve, or a basket sized by ex-ante covariance rather than equal weights.

13. Limitations

1. *Historical data only.* A backtest describes the past; the regime mix of the sample shapes every conclusion.
2. *Backward-looking volatility.* The risk estimate reacts to realized turbulence rather than forecasting shocks, contributing directly to the COVID failure.

3. *Monthly signal latency.* The rebalance cadence is the root cause of the fast-crash vulnerability; a faster signal is the obvious next experiment.
4. *Simplified frictions.* Turnover-based costs are charged, but bid-ask spreads, slippage, taxes, and market impact are not separately modeled.
5. *Compact universe.* Four ETFs aid interpretability but limit diversification.

Relative to the initial implementation, three earlier limitations are now resolved: lookahead bias is eliminated, out-of-sample/walk-forward validation and significance testing are included, and risk-adjusted metrics use a real risk-free rate.

14. Conclusion

This project treats financial markets as a quantitative systems problem: broad ideas about momentum and risk are translated into explicit mathematical rules, implemented without common backtest errors, and judged against falsifiable hypotheses. Its contribution is not a winning trading system, and it does not claim to be one. It is a demonstration that a strategy can be specified precisely, tested rigorously, and reported honestly enough to reveal both what works and what does not.

The evidence is mixed in an informative way. The strategy delivers genuine downside protection versus pure equity (H1 supported) but does not outperform a balanced benchmark, and its protective benefits are not statistically significant in a full-sample test (H4: only the return-sacrifice versus equity is significant unconditionally). Conditioning on down markets sharpens this: the strategy loses significantly less than equity in stress months, and the engineered crash-guard fix (v3) extends that significant protection to the balanced benchmark as well while sharply reducing fast-crash drawdowns, so its value is real and regime-conditional (H2 supported once conditioned). Its character is stable out-of-sample and across the parameter grid, where no configuration beats a 60/40 (H3 supported as a robust-underperformance result); a walk-forward test shows that optimizing parameters makes things worse, not better; and the v3 fix reveals that crash protection and whipsaw are inseparable ends of a single speed dial, the project's deepest finding, reachable only by diagnosing a weakness and engineering against it. The shift this work represents (from viewing markets as a domain of instinct to seeing them as systems that can be modeled, stress-tested, fixed, and falsified) is what made it most valuable to me, and it is the lens I intend to bring to quantitative finance.

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Data: Yahoo Finance via the `yfinance` library. *Implementation:* `numpy`, `pandas`, `scipy`, `matplotlib`.

Appendix A: Reproducibility

All results are generated by `quant_momentum_ALL_IN_ONE.py`. Price and risk-free data are cached to CSV; the bootstrap uses a fixed seed (42). The repository contains the script, the cached data, the generated figures, and this document, so any reader can reproduce every number. Sample span: 2012-01-03 to 2026-06-15 (112 month-end rebalances; backtest active from 2013-02-01).

Appendix B: Parameter Grid

Parameter	Baseline	Grid tested
Momentum lookback L	252	63, 126, 252, 504
Skip s	0	0 (baseline), 21 (extension)
Volatility window K	20	20, 60, 90
Target volatility σ^*	10%	5%, 10%, 15%
Max weight $w_{\max}$	1.0	-
Transaction cost κ	0.1%	0.0%, 0.05%, 0.1%, 0.2%
Risk-on universe	VOO, VXUS	-
Risk-off asset	BND	BND, SHY, TLT